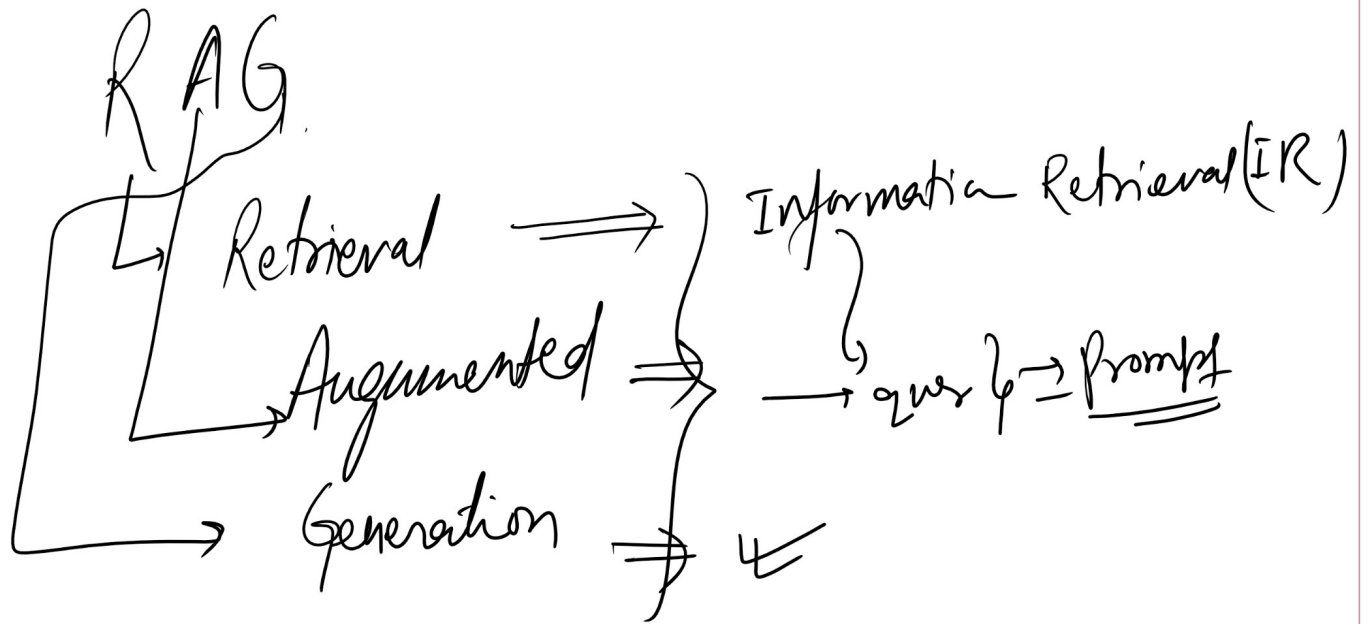
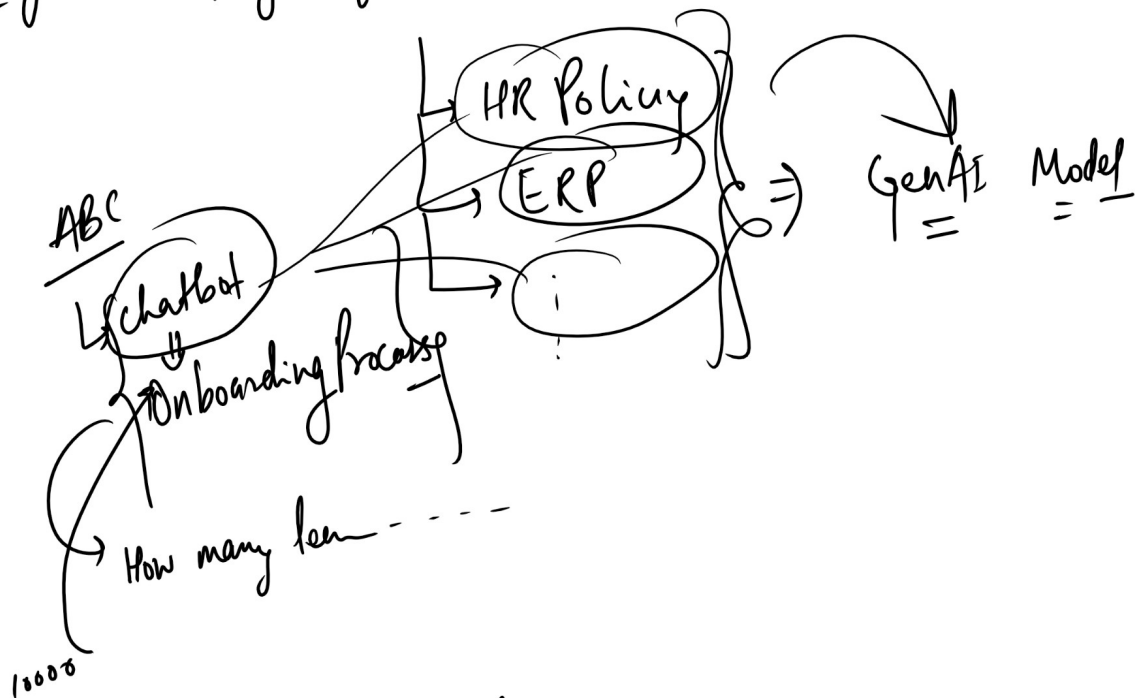




Why :- Company Confidential doc. s



→ chatGPT ⇒ what is my mother's name?

→ How many team s team ?

⇒ medical Domain ⇒ very specific

[llama 3.1 8b] MRI

llama 3.1 & b → Knowledge Cutoff ⇒ Dec 2023
 hallucinate
 ↓
 mitigate ⇒ RAG

RAG ⇒ 1. Data Ingestion Pipeline
 2. Augment & Generation Pipeline

1. pdf ⇒ Maaharvat ⇒ [] ⇒ 128K ⇒ llama 3.1 8b
 If I process the entire document at once

⊙ slow

⊙ data loss ⇒ Hallucination

pdf
 docx
 sgi
 week
 htm
 Doc.

langchain loaders ⇒

LangChain Document Structure

LangChain Document

from langchain.schema import Document

Core Components:

- page_content (str)
- metadata (dict)

```

# Creating a document
doc = Document(
    page_content="RAG is a technique...",
    metadata={
        "source": "chapter1.pdf",
        "page": 1,
        "timestamp": "2024-01-15"
    }
)
  
```

page_content (String)

The actual text content of the document

- Contains the main information to be embedded and searched
- Must be a string (can be any length)

Examples:

Research Paper:

"Retrieval-Augmented Generation (RAG) combines the benefits of pre-trained language models with information retrieval systems to generate more accurate and contextual responses."

Product Manual:

"To install the software, first ensure your system meets the minimum requirements: Windows 10 or later, 8GB RAM, and at least 20GB of free disk space."

Best Practices:

- Keep content focused and coherent

metadata (Dictionary)

Additional information about the document

- Used for filtering, tracking, and context
- Can contain any JSON-serializable data

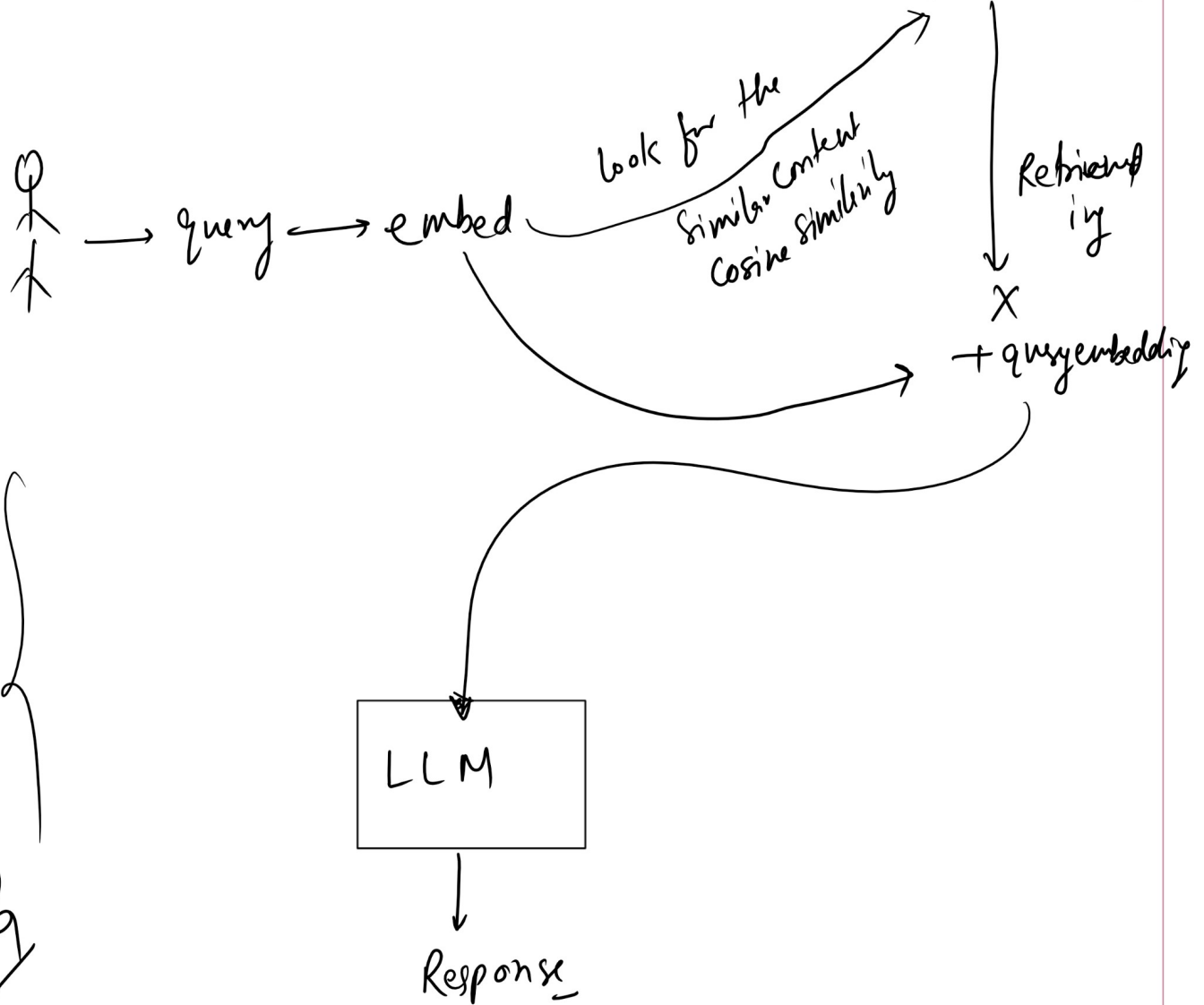
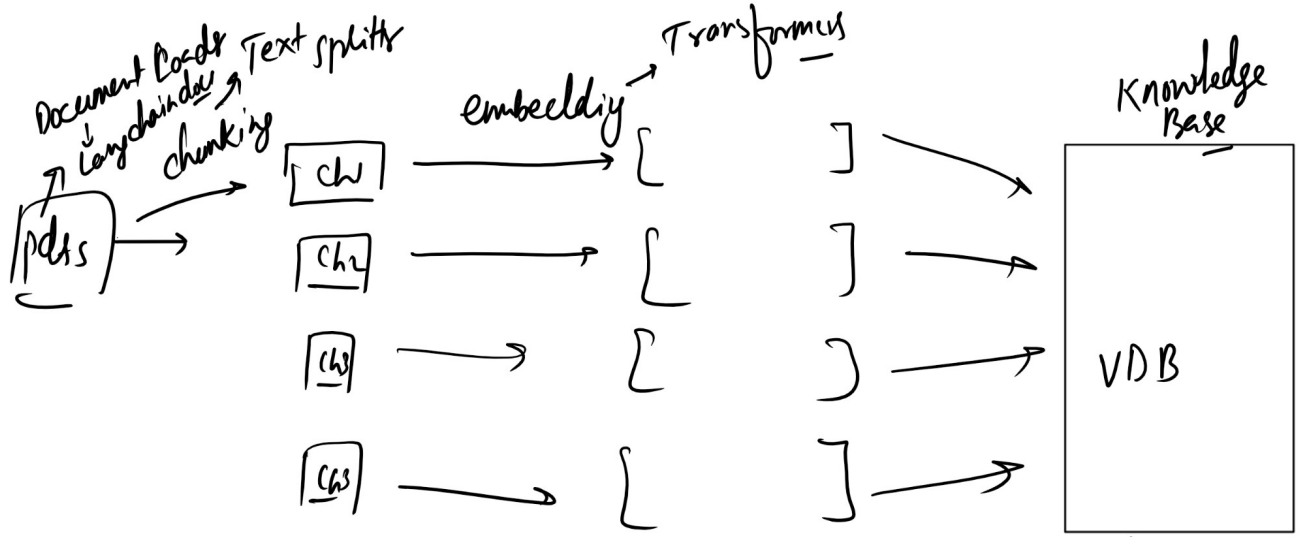
Common Metadata Fields:

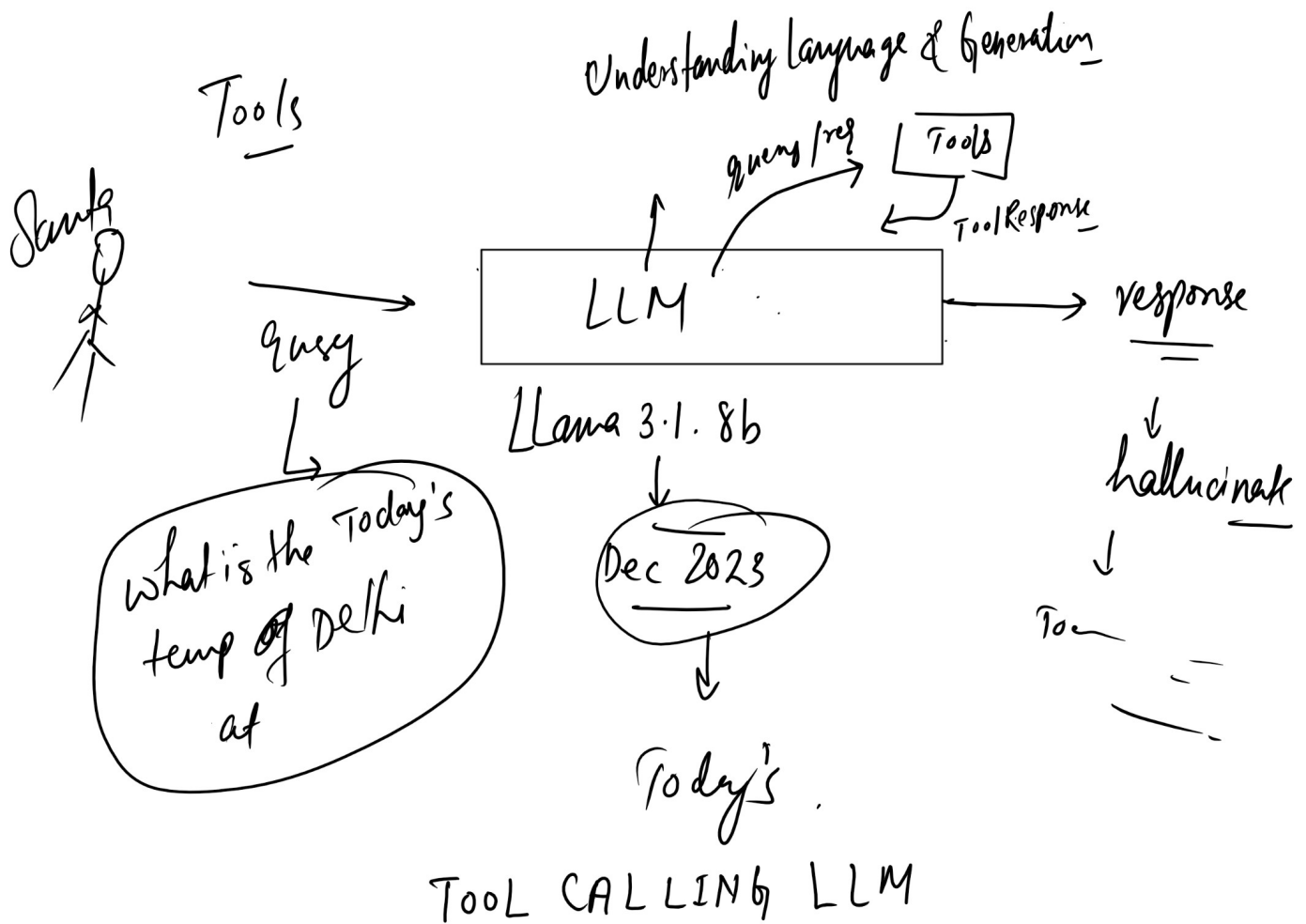
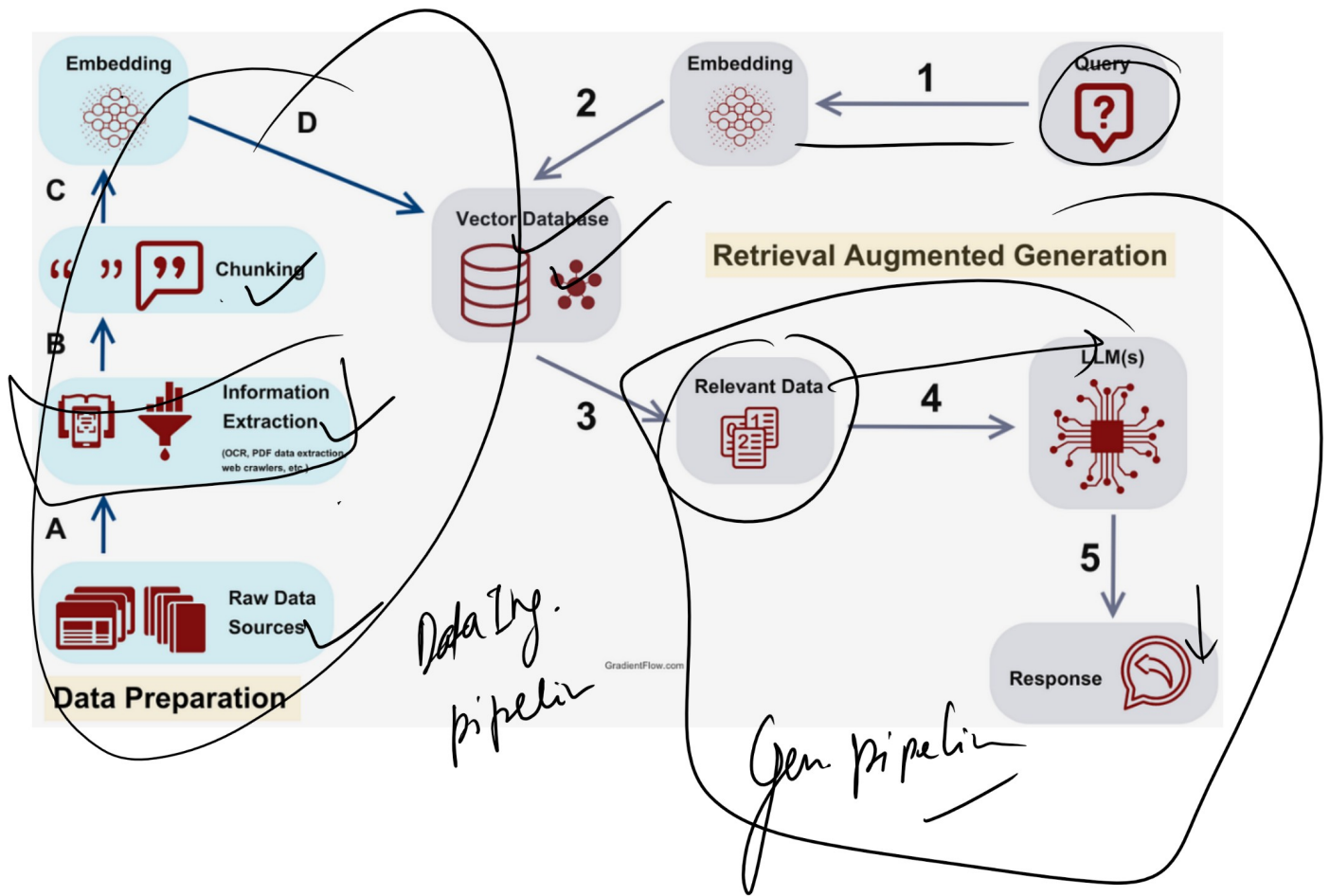
source File path or URL "docs/manual.pdf"	page / chunk_id Location in document page: 42, chunk: 7
timestamp Creation/modification date "2024-01-15T10:30:00Z"	author Document creator "John Doe"
category / type Document classification "technical", "legal"	language Content language "en", "es", "fr"

1. Page Content

2. Metadata

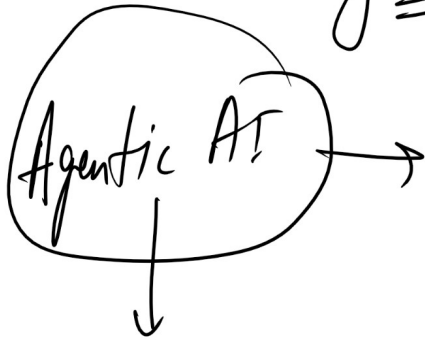
dictionary





↓ prompt → As Instruction

Agent ₌ ⇒ Standalone software



Goal

↳ Perceiving
↳ Reasoning
↳ Action
↳ Learning